Sequences and discrete models

Lecture and worked examples

Chapter path

Objectives. Turn a story into a sequence; distinguish increment from rate; compute terms and sums; justify monotonicity; investigate a threshold.

Prerequisites. Algebraic manipulation, integer powers, percentages, affine function, Python loop.

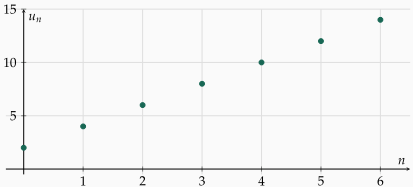
Yearly reference point. 8-17 September: 8 hours, then revisits

Defining and representing a sequence

A real sequence is a function whose variable is an integer: to each allowed index n corresponds a real number u_n. The notation (u_n) denotes the sequence, u_n a term. An explicit formula gives u_n directly in terms of n. A definition by recurrence gives a first term and a rule for passing from one term to the next. The representation is a set of points (n, u_n): these points are not necessarily joined up.

Defining and representing a sequence

For $u_n = n^2 + 1$, $u_0 = 1$, $u_3 = 10$. For $v_0 = 1$ and $v_{n+1} = 2v_n + 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. We cannot replace v_3 by $2 \times 3 + 1$: the rule applies to the previous term.



Defining and representing a sequence

The index is not the value. A finite list of terms does not determine a unique rule.

Defining and representing a sequence

Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; suggest an explicit formula.

Arithmetic sequences

A sequence is arithmetic with common difference r if $u_{n+1} = u_n + r$ for every allowed index. Then $u_n = u_p + (n - p)r$ for $n \geq p$. If $r > 0$ it is strictly increasing; if $r < 0$, strictly decreasing; if $r = 0$, constant. The points lie on a straight line.

Arithmetic sequences

A tank holds 140 L and receives 12 L per minute. After n minutes, $u_n = 140 + 12n$. At 5 min: 200 L. If we start the indexing at 1 with 140 L, we write $u_n = 140 + 12(n - 1)$.

Arithmetic sequences

Adding a constant percentage does not generally give an arithmetic sequence.

Arithmetic sequences

We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine r and u_0.

Geometric sequences

A sequence is geometric with common ratio q if $u_{n+1} = qu_n$. For $n \geq p$, $u_n = u_p q^{n-p}$, with the convention that an empty product equals 1. A constant rate t gives $q = 1 + t$. Monotonicity is determined using $u_{n+1} - u_n = (q-1)u_n$. For $q < 0$, the signs may alternate.

Geometric sequences

A quantity of 160 mg loses 25% at each step: $u_n = 160(\frac{3}{4})^n$. The terms are positive and $u_{n+1} - u_n = -\frac{u_n}{4} < 0$. We obtain $u_2 = 90$ mg, without confusing a 25% decrease with removing 25 mg.

Geometric sequences

$q > 1$ is not enough to conclude that every geometric sequence is increasing: the sign of the first term matters.

Geometric sequences

Study the monotonicity of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$.

Sums, indices and proofs

For $n \geq 1$, $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$. For $q \neq 1$ and $n \geq 0$, $1 + q + \cdots + q^n = \dfrac{1 - q^{n+1}}{1 - q}$. For $q = 1$, the sum equals $n + 1$. Hence, from u_p to u_n, an arithmetic sum equals $\dfrac{(n - p + 1)(u_p + u_n)}{2}$; a geometric sum equals $\dfrac{u_p(1 - q^{n-p+1})}{1 - q}$ if $q \neq 1$.

For $u_n = 3 + 2n$, $u_2 + \cdots + u_6 = \dfrac{5(7+15)}{2} = 55$. For $v_n = 2 \times 3^n$, $v_0 + \cdots + v_3 = \dfrac{2(1-3^4)}{1-3} = 80$.

Sums, indices and proofs

Count the terms before using the formula: from 0 to n there are n+1.

Calculate $\sum_{k=2}^{5} 2^k$ without the summation symbol in the answer.

Conjecturing a limit and programming a threshold

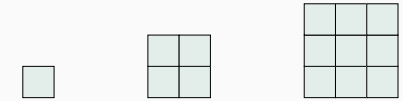
A limit is approached here intuitively. The terms $\frac{1}{n+1}$ get closer to 0; n^2 grows without bound; $(-1)^n$ does not settle. A numerical experiment suggests a conclusion but does not prove a behaviour for all indices. A threshold algorithm must specify the direction of the inequality and provide a stopping condition.

Conjecturing a limit and programming a threshold

For $u_n = 80(\frac{1}{2})^n$, the first index with $u_n < 6$ is 4: $u_3 = 10 \geq 6$ and $u_4 = 5 < 6$. The program starts with $n = 0, u = 80$ and repeats the division by 2 while $u \geq 6$.

Conjecturing a limit and programming a threshold

A loop with no bound may never stop if the threshold cannot be reached; $<$ and $\leq$ are not interchangeable.

Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$.